

Interest rate derivatives
are more difficult to value
than equity derivatives

- behaviour of int rate = more complicated than a stock
- valuation of many products, necessary to develop a model describing the entire yield curve
- volatilities of different points on the curve are different
- int rates used for $\left\{ \begin{array}{l} \text{discounting} \\ \text{determining a payoff} \end{array} \right.$

call:

$$c = P(0,T) [F_0 \tilde{\Phi}(d_1) - K \tilde{\Phi}(d_2)]$$

put:

$$p = P(0,T) [K \tilde{\Phi}(-d_2) - F_0 \tilde{\Phi}(-d_1)]$$

$$d_1 = \frac{\ln(\frac{F_0}{K}) + (\frac{\sigma^2 T}{2})}{\sigma \sqrt{T}} \quad d_2 = \frac{\ln(\frac{F_0}{K}) - (\frac{\sigma^2 T}{2})}{\sigma \sqrt{T}} = d_1 - \sigma \sqrt{T}$$

③ σ = volatility of F

Approximations:

- $E(V_T)$ = forward price F_0
- stochastic behaviour of int rates is not taken into account in the way discounting is done

Pricing bond options:

$$F_0 = \frac{B_0 - I}{P(0,T)} \rightarrow \text{bond price at } t=0 \text{ PV of coupons paid during bonds lifetime}$$

Price + yield volatilities:

$$\sigma = \rho y_0 \sigma_y \rightarrow \text{initial forward yield + yield volatility}$$

$$D = \frac{\text{duration of bond after the option}}{(1 + \frac{1}{m})}$$

Interest Rate CAP

provides insurance against risk of floating rate reaching above a certain level $\rightarrow R_x = \text{cap rate}$

$$\text{caplet payoff} = L \times 0.25 \times \max(R_h - R_x, 0)$$

$$= L \delta_K \max(R_h - R_x, 0) \rightarrow R_h \text{ reset every 3 months = Tenor (time between resets)}$$

R_h = floating rate
 R_x = cap

compounding frequency = reset

call:

$$c = [\delta_K P(0, t_{m+1}) (F_K \tilde{\Phi}(d_1) - R_x \tilde{\Phi}(d_2))]$$

③ value each caplet separately

Interest Rate FLOOR

put:

$$p = [\delta_K P(0, t_{m+1}) (R_x \tilde{\Phi}(-d_2) - F_K \tilde{\Phi}(-d_1))]$$

③ All valued at $(t_K) \rightarrow$ start of period

cap price = floor price + value of swap

④ put-call parity

receive floating + pay fixed

F_0, B_0, X
 $\rightarrow$ Dirty Price:
Price paid + income

(CHAPTER 13:
VALUATION OF
INVESTMENTS

Credit default swap

THEORY: yield on bond > RFR

PRACTICE: Price CDS - (yield > RFR) $\neq 0 \rightarrow$ volatile

The above reflects:

1. package of bond + CDS, to protect against default is not risk-free in practice
 - counterparty risk on CDS
 - No default value may be $\uparrow$ or $\downarrow$ face value
 - documentation differences
2. Package of bond + CDS = illiquid + requires funding
 - basis = 0ive when funding is expensive
3. Supply + demand dynamics differ in different markets

③ Risk-neutral prob default = $\tilde{\Phi}(-d_2)$

Floor + caps as portfolio of bond prices

cap $\rightarrow$ put option on $\tilde{z}_{CB}$

floor $\rightarrow$ call option on $\tilde{z}_{CB}$

1. spot volatilities - different σ_i for each cap
2. flat volatilities - same average σ for all caps

Swaptions

1. Pay fixed, Receive floating $\rightarrow$ CALL option

$$\frac{L}{m} P(0, t_i) [F_0 \tilde{\Phi}(d_1) - R_x \tilde{\Phi}(d_2)] \rightarrow \# \text{ of CF each year + compounding frequency}$$

$$\sum_{i=1}^n \frac{L}{m} P(0, t_i) [F_0 \tilde{\Phi}(d_1) - R_x \tilde{\Phi}(d_2)] = A$$

$$\therefore = LA [F_0 \tilde{\Phi}(d_1) - R_x \tilde{\Phi}(d_2)]$$

2. Pay floating, Receive fixed $\rightarrow$ PUT option

$$\frac{L}{m} P(0, t_i) [R_x \tilde{\Phi}(d_2) - F_0 \tilde{\Phi}(-d_1)]$$

$$\therefore = LA [R_x \tilde{\Phi}(d_2) - F_0 \tilde{\Phi}(-d_1)]$$

Pricing: Merton model

V_t = value of assets at time t
 D = debt due to be paid at T
 E_t = value of equity at time t

$$V_t \leq D \rightarrow \text{default debt } E_T = 0$$

$$V_t > D \rightarrow \text{debt repayment } E_T = V_T - D$$

$$\therefore E_t = \max(0; V_t - D)$$

$$E_0 = V_0 \tilde{\Phi}(d_1) - D e^{-rT} \tilde{\Phi}(d_2)$$

Evaluating securitisation

Assets for securitisation focus on $\begin{cases} \text{predictability} \\ \text{sustainability} \end{cases}$

consider:

- lease terms
- rental prospects
- degree of potential competition
- barriers to competitive entry

⊗ Discounted CF approach

modelling features

- statistics → early repayment experience
- probability → default / recovery + timing
- treasury management → payments in/out
- structuring + security issuance

⊗ Difficulty = what interest rate

of CF

→ sufficient to pay holders of asset-backed securities